

DATA STRUCTURES (CI33)

Practice Questions

	Unit 1
1	Differentiate between a normal variable and a pointer variable.
2	What is the use of the * (<i>asterisk</i>) and & (<i>ampersand</i>) operators in pointers? Illustrate with a simple example C program.
3	Write a program to find the sum of elements in an array using pointers.
4	Write a C program to swap two numbers using pointers and function.
5	Explain pointer arithmetic with examples.
6	State the purpose of following types of pointers. Illustrate each with few lines of syntax. i. Null pointer ii. Dangling pointer iii. Void pointer iv. Wild pointer v. Double pointer vi. Function pointer
7	What is dynamic memory allocation? How is it different from static memory allocation?
8	Explain the four dynamic memory allocation functions in C.
9	Write a short program to allocate memory dynamically for an array using <i>malloc()</i> .
10	Draw taxonomy of various Data Structures.
	Unit 2
1	Elaborate with an algorithm or (C-program) to convert a given infix expression to postfix expression.
2	Convert the following infix expressions to postfix expression. Show the tracing. i) $A + B * C - (D/E^F) * G - H$ ii) $A * (B - C/D) + G \% F$ iii) $p - q/r * (s^t + u)$ iv) $m + n ^ p \% q * (u - v/w)$
3	Write an algorithm for evaluating a valid postfix expression.
4	Evaluate the following postfix expressions by showing tracing of stacks. i) $4\ 5\ 6\ *\ +\ 2\ 4\ 3\ ^\ /\ -$ ii) $6\ 5\ 4\ *\ 2\ /\ +\ 7\ 3\ ^\ -$ iii) $a\ b\ /\ c\ *\ d\ e\ ^\ -\ f\ +$ Here $a = 12, b = 3, c = 5, d = 15, e = 4, f = 20$

5	Write an algorithm to convert a given infix expression to prefix expression. Convert the following to prefix expressions. i. $a + b/c - d * e$ ii. $a * (b + c \% d) - e/f$
6	Define a recursion. List the important features of a recursive program.
7	Discuss the recursive and iterative methods to solve Tower of Hanoi problem.
8	Evaluate the following prefix expressions by showing tracing of stacks. i) $+ / 4 7 * 5 2$ ii) $+ / 8^4 3 * 9 5$
9	Give classification of queue data structure. List the applications of queues.
10	What is circular queue? Write the C implementation of circular queue using array.

Note: Students should maintain a separate notebook and write the answers neatly. This will aid in theory exams. Also its evaluated as part of CIE.